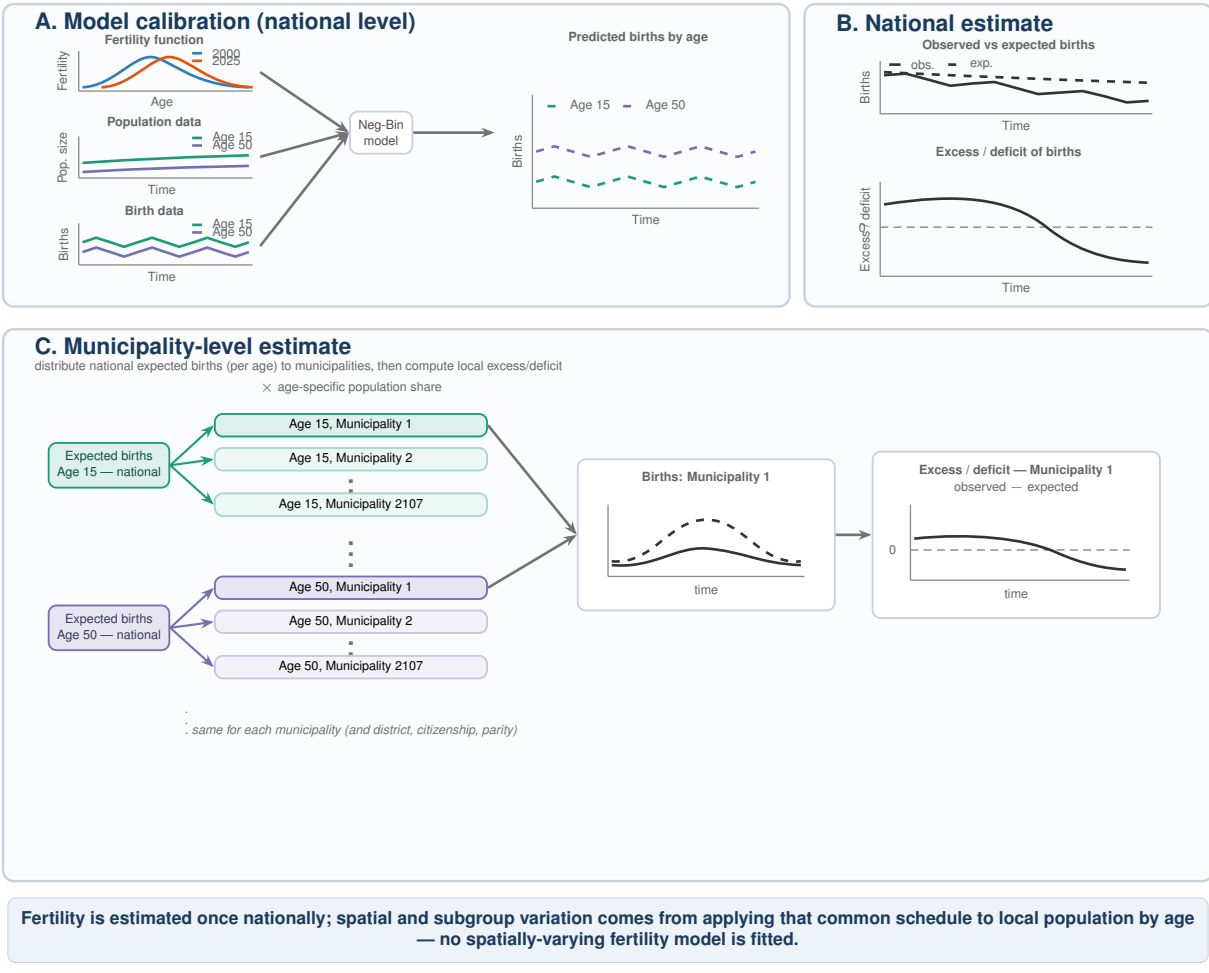
A. Model calibration (national level)
Fertility function
Fertility
2000
2025
Age
Population data
Pop. size
Age 15
Age 50
Time
Birth data
Births
Age 15
Age 50
Time
Neg-Bin model
Predicted births by age
Births
Age 15
Age 50
Time
B. National estimate
Observed vs expected births
obs.
exp.
Births
Time
Excess / deficit of births
Excess / deficit
Time
C. Municipality-level estimate
distribute national expected births (per age) to municipalities, then compute local excess/deficit
× age-specific population share
Expected births Age 15 — national
Age 15, Municipality 1
Age 15, Municipality 2
Age 15, Municipality 2107
Expected births Age 50 — national
Age 50, Municipality 1
Age 50, Municipality 2
Age 50, Municipality 2107
same for each municipality (and district, citizenship, parity)
Births: Municipality 1
time
Excess / deficit — Municipality 1
observed — expected
0
time
Fertility is estimated once nationally; spatial and subgroup variation comes from applying that common schedule to local population by age — no spatially-varying fertility model is fitted.